

Plasma concentrations of VCAM-1 and PAI-1: A predictive biomarker for post-operative recurrence in colorectal cancer

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This prospective study used antibody suspension bead arrays to identify biomarkers capable of predicting post-operative recurrence with distal metastasis in patients with colorectal cancer. One hundred colorectal cancer patients who underwent surgery were enrolled in this study. The median follow-up period was 3.9 years. The pre-operative plasma concentrations of 24 angiogenesis-related molecules were analyzed with regard to the TNM stage and the development of post-operative recurrence. The concentrations of half of the examined molecules (13/24) increased significantly according to the TNM stage ($P < 0.05$). Meanwhile, a multivariate logistic regression analysis revealed that the concentrations of vascular cell adhesion molecule 1 (VCAM-1) and plasminogen activator inhibitor-1 (PAI-1) were significantly higher in the post-operative recurrence group. The VCAM-1 and PAI-1 model discriminated post-operative recurrence with an area under the curve of 0.82, a sensitivity of 0.75, and a specificity of 0.73. A leave-one-out cross-validation was applied to the model to assess the prediction performance, and the result indicated that the cross-validated error rate was 12.5% (12/96). In conclusion, our results demonstrate that antibody suspension bead arrays are a powerful tool to screen biomarkers in the clinical setting, and the plasma levels of VCAM-1 and PAI-1 together may be a promising biomarker for predicting post-operative recurrence in patients with colorectal cancer. (*Cancer Sci* 2010; 101: 1886–1890)

Colorectal cancer (CRC) is one of the leading causes of death in Japan (<http://ganjoho.ncc.go.jp/public/statistics/index.html>) and Western countries.⁽¹⁾ Despite recent advances in adjuvant chemotherapy and surgical techniques, 20–40% of patients die because of metastasis after curative surgery.⁽²⁾ Tumor-node-metastasis (TNM) staging is well established and the most reliable system for predicting the outcome of CRC. In particular, the TNM staging system works very well for predicting the outcome of early stage I cancers and advanced stage IV cancers. However, the 5-year survival rate varies from 44% to 83% within TNM stage III, indicating that a wide variation in outcomes exists within each stage as a result of biological heterogeneity.⁽³⁾ Thus, highly accurate predictors of post-operative recurrence are needed for patients with CRC who undergo curative surgery, as such predictors would likely contribute to the further improvement of the 5-year survival rate by justifying the addition of intensive adjuvant chemotherapy to the therapeutic regimens of subgroups with a high risk of post-operative recurrence. Therefore, the prediction of post-operative recurrence is regarded as one of the most important research themes in clinical settings and has been extensively studied, with particular attention given to the investigation of various molecular prognostic factors.

In addition to the TNM stage, the carcinoembryonic antigen (CEA) level is routinely used to monitor recurrence in patients with CRC.⁽⁴⁾ A large clinical study demonstrated that pre-operative CEA levels provide prognostic information in addition to that provided by the TNM staging system and determined that the pre-operative CEA level was an independent predictor of survival and recurrence.⁽⁵⁾ However, the study concluded that although an elevated pre-operative CEA level (>5 mg/mL) may be correlated with a poor prognosis, the available data was insufficient to support the use of the CEA level for determining whether a patient should undergo adjuvant therapy.⁽⁴⁾ Other molecular markers, including the K-Ras mutation status,^(6,7) microsatellite instability,⁽⁸⁾ the loss of heterogeneity at 18q,⁽⁹⁾ and p53,⁽¹⁰⁾ have been examined with regard to predicting the outcome of subgroups; unfortunately, none of these molecular markers are suitable for routine clinical use. Thus, further investigations of novel molecular markers are eagerly awaited.

The Bio-Plex suspension array system (Bio-Rad Laboratories, Hercules, CA, USA) utilizes a series of color-coded beads, each of which is coupled to a unique antibody specific for a biochemical marker. This assay is capable of measuring the levels of multiple targets in a single well of a 96-well microplate using as little as 12.5 μ L of serum, plasma, or other matrix. In the present study, 24 angiogenesis-related markers from this assay panel were used to evaluate plasma proteins and their potential associations with disease progression and the recurrence of CRC.

Materials and Methods

Patient selection. Patients with histologically confirmed colorectal cancer who were between the ages of 20 and 80 years and who were scheduled to undergo surgery were eligible for enrollment in this study. Additional inclusion criteria included an Eastern Cooperative Oncology Group performance status of 0–2. All the patients in this series underwent surgery. This prospective study was approved by the Institutional Review Board of the National Cancer Center Hospital and written informed consent was obtained from all the patients.

Clinical and pathologic features. Clinical features including age, sex, primary site of tumor, histologic type of tumor, TNM stage, and post-operative recurrence were recorded. A pathologist reviewed the microscopic slides. Post-operative recurrence was defined when distant metastasis was observed. The median follow-up period of post-operative recurrence was 3.9 years.

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Preparation of plasma samples. Two milliliters of whole blood were collected into EDTA-containing tubes before surgery (within 2 weeks) and were centrifuged at 1500 *g* for 10 min to obtain the plasma samples. The samples were stored at -80°C until further use.

Angiogenesis-related molecules. The 24 plasma markers used in this study were as follows: interleukin 6 receptor (IL-6R), matrix metalloproteinase 9 (MMP-9), TIMP metalloproteinase inhibitor 1 (TIMP-1), TIMP metalloproteinase inhibitor 2 (TIMP-2), endostatin, P-selectin, intercellular adhesion molecule 1 (ICAM-1), vascular cell adhesion molecule 1 (VCAM-1), Tie-2, plasminogen activator inhibitor-1 (PAI-1), macrophage migration inhibitory factor (MIF), plasminogen activator urokinase receptor (uPAR), angiopoietin 2 (Ang-2), follistatin, hepatocyte growth factor (HGF), interleukin 8 (IL-8), colony stimulating factor 3 (G-CSF), platelet-derived growth factor beta polypeptide (PDGF-BB), vascular endothelial growth factor (VEGF), leptin, platelet/endothelial cell adhesion molecule (PECAM-1), interleukin 12 (IL-12), fibroblast growth factor 2 (FGF-basic), and tumor necrosis factor (TNF- α). Ang-2, Follistatin, The nine markers of HGF, IL-8, PDGF-BB, VEGF, Leptin, PECAM-1, and G-CSF are commercially available as the Human Premixed Angiogenesis (9) panel (Bio-Rad Laboratories). The others are available for customization or in developing markers.

Antibody suspension bead arrays system. The plasma concentrations of each molecule were measured using a Bio-Plex suspension array system (Bio-Rad Laboratories), which permits the simultaneous measurement of multiple circulating proteins in a single well using only 12.5 μL of plasma. The assay was performed according to the manufacturer's instructions and a previously described method.⁽¹¹⁾ All plasma samples were diluted 1 in 4 with the appropriate diluents prior to assay. The samples were tested in duplicate.

Statistical analysis. The correlations between the plasma concentrations and the TNM stages were analyzed using a linear trend analysis and the proportional odds model. In the linear trend analysis, we used a one-way ANOVA model with a linear contrast, which consisted of the TMN stage scores. A *t*-test was used to compare the post-operative recurrence and the no recurrence groups. In the multivariate analysis, an analysis was performed for all the plasma markers but not for the clinical variables because too many explanatory variables for the sample size were included in this study. A logistic regression analysis was used to examine statistical differences according to post-operative recurrence and a stepwise method was used to select the most useful explanatory parameters. Analyses were performed using SAS software version 9.1.3 (SAS Institute, Cary, NC, USA). A *P*-value of <0.05 was considered statistically significant.

Results

Patient results. A total of 100 consecutive patients were enrolled in this study. The median age of the enrolled patients was 59 years (range, 31–79 years). Among them, 96 patients received curative operations and four patients had distal metastasis and received palliative operations. Eleven patients developed recurrences with distal metastasis during the follow-up period. Table 1 summarizes the characteristics of the patients and their tumors.

Plasma concentrations of 24 angiogenesis-related molecules. We measured the plasma concentrations of 24 angiogenesis-related molecules: IL-6R, MMP-9, TIMP-1, TIMP-2, endostatin, P-selectin, ICAM-1, VCAM-1, Tie-2, PAI-1, MIF, uPAR, Ang-2, follistatin, HGF, IL-8, G-CSF, PDGF-BB, VEGF, leptin, PECAM-1, IL-12, FGF-basic, and TNF- α (Table 2). Overall, 98.5% of the plasma samples were successfully quantified using a standard curve.

Table 1. Patient characteristics

Characteristics	Curative ope.		Palliative ope.	Total
	Rec+	Rec–		
Age (years)				
≥60	6	42	2	50
<60	5	43	2	50
Sex				
Male	3	54	3	65
Female	8	31	1	35
Primary site				
Colon	8	40	1	49
Rectum	3	45	3	51
Hist. type				
Well diff.	8	59	3	70
Others	3	26	1	30
TNM stage				
I	1	26	–	27
II	2	22	–	24
III	8	37	–	45
IV	–	–	4	4
Total	11	85	4	100

Hist. type, histology of primary tumor; Rec+, post-operative recurrence (+); Rec–, post-operative recurrence (–).

Tumor-node-metastasis (TNM) stage and plasma concentrations of angiogenesis-related molecules. The TNM stage can be accurately used to stratify patients at a high risk for cancer progression and is thought to reflect the malignant potential of each tumor. To estimate the contributions of the angiogenesis-related molecules to the malignant potentials of the tumors, we examined the correlation between the plasma concentrations of each molecule and the TNM stage. A linear trend analysis showed that the plasma concentrations of 13 molecules increased significantly with an increasing TNM stage ($P < 0.05$): IL-6R, TIMP-1, TIMP-2, P-selectin, Tie-2, PAI-1, uPAR, Ang-2, follistatin, HGF, IL-8, PDGF-BB, and VEGF (Table 3). Next, we performed an exploratory multivariate analysis using a proportional odds model with the TNM stage (I–IV) assigned as the objective variable and each of the angiogenesis-related molecules assigned as explanatory variables. The multivariate analysis identified TIMP-1, P-selectin, Ang-2, HGF, IL-8, PDGF-BB, and VEGF as being significantly correlated with the TNM stage. These results indicated that the plasma concentrations of several molecules increased significantly with an increasing TNM stage, strongly suggesting that these molecules might be candidate biomarkers for an unfavorable outcome in patients with CRC.

Post-operative recurrence and plasma concentrations of angiogenesis-related molecules. To predict post-operative recurrence using this system, we analyzed the 96 patients with CRC who underwent curative operations, excluding the four patients with distal metastasis. Among these 96 patients, 11 developed recurrences during the follow-up period; the remaining 85 patients did not show any signs of recurrence. When the plasma levels of the angiogenesis-related molecules were compared between the patients with recurrences and those without recurrences, *t*-tests demonstrated that the plasma concentrations of IL-6R (63.2 ± 13.8 and 51.3 ± 17.0 ng/mL, respectively), P-selectin (76.1 ± 24.4 and 60.0 ± 24.8 ng/mL, respectively), VCAM-1 (163.9 ± 61.0 and 134.6 ± 35.0 ng/mL, respectively), and PAI-1 (28.2 ± 15.4 and 19.8 ± 10.2 ng/mL, respectively) were significantly higher among the patients with recurrences (Fig. 1a–d, Table 4). A multivariate logistic regression analysis revealed that the plasma concentrations of VCAM-1 and PAI-1 were significantly higher among the patients with recurrences ($P = 0.039$ and $P = 0.028$, respectively). A stepwise

Table 2. Plasma concentrations of 24 angiogenesis-related molecules in 100 colorectal cancers

Molecules	Range	Average (SD)	25%	Percentile	
	pg/mL			Median	75%
IL-6R	23–149	54 (18)	42	52	62
MMP-9	6–189	35 (27)	20	27	41
TIMP-1	44–283	117 (35)	96	118	135
TIMP-2	9–47	24 (5)	21	24	28
Endostatin	90–456	187 (64)	140	177	223
P-selectin	0–164	64 (27)	48	62	78
ICAM-1	103–605	282 (81)	225	270	317
VCAM-1	76–333	138 (40)	110	135	163
Tie-2	9–1070	143 (164)	48	73	182
PAI-1	3–65	21 (12)	14	19	26
MIF	0–120	53 (26)	44	59	69
uPAR	1–130	24 (23)	8	15	37
Ang-2	0–6147	1381 (1337)	415	938	2197
Follistatin	235–2903	924 (614)	502	686	1197
HGF	201–12213	2700 (2516)	1076	1628	3930
IL-8	5–234	52 (41)	24	34	74
G-CSF	0–4775	832 (1133)	79	247	1161
PDGF-BB	6–6219	737 (831)	220	439	922
VEGF	0–724	186 (164)	67	120	262
Leptin	0–32847	3155 (4433)	1149	2134	3851
PECAM-1	1188–15837	5562 (3472)	2901	4487	7348
IL-12	0–32	5 (6)	2	3	6
FGF-basic	0–235	21 (30)	4	13	25
TNF- α	0–72	4 (9)	1	2	4

The concentrations are ng/mL for interleukin 6 receptor (IL-6R), matrix metalloproteinase 9 (MMP-9), TIMP metalloproteinase inhibitor 1 (TIMP-1), TIMP-2, endostatin, P-selectin, intercellular adhesion molecule 1 (ICAM-1), vascular cell adhesion molecule 1 (VCAM-1), Tie-2, plasminogen activator inhibitor-1 (PAI-1), macrophage migration inhibitory factor (MIF), and plasminogen activator urokinase receptor (uPAR), and the others are pg/mL. FGF-basic, fibroblast growth factor 2; G-CSF, colony stimulating factor 3; HGF, hepatocyte growth factor; PDGF-BB, platelet-derived growth factor beta polypeptide; PECAM-1, platelet/endothelial cell adhesion molecule; TNF- α , tumor necrosis factor; VEGF, vascular endothelial growth factor.

method selected VCAM-1 and PAI-1 as the most useful explanatory parameters, suggesting that the combination of these two molecules might synergistically improve the prediction of post-operative recurrence. Finally, a prediction model incorporating VCAM-1 and PAI-1 successfully discriminated post-operative recurrence, with an area under the curve (AUC) of 0.82, a sensitivity of 0.75, and a specificity of 0.73 (Fig. 2a). To assess the prediction performance, a leave-one-out cross-validation was applied to the model. The cross-validated error rate was 12.5% (12/96). In stage III patients, the prediction model had a sensitivity of 0.625 (5/8) and a specificity of 0.865 (32/37) for predicting post-operative recurrence (Fig. S2). On the other hand, when apparent distal metastases of CRCs were applied to the VCAM-1/PAI-1 prediction model, three out of four metastatic cases were determined as “recurrence (+) cases”. These results suggest that apparent metastatic tumors could be discriminated using the model. Although a validation study is necessary, our results raise the possibility that the combined use of the pre-operative plasma concentrations of VCAM-1 and PAI-1 might be useful for predicting post-operative recurrence in patients with CRC. Finally, we retrospectively analyzed the plasma PAI-1 concentrations of metastatic and non-metastatic CRC in another study of an independent cohort using conventional ELISA. The plasma concentrations in the metastatic CRC patients were significantly higher than those in the non-metastatic patients ($P = 0.005$), even in the independent cohort (Fig. S1). The CEA level was not significantly different between

Table 3. Tumor-node-metastasis (TNM) stage and plasma concentrations in 100 colorectal cancers

Molecules	TNM stage				Univariate	Multivariate
	I	II	III	IV	P-value	P-value
IL-6R	50	57	52	76	0.01	n.s.
MMP-9	34	50	28	33	n.s.	n.s.
TIMP-1	107	122	114	185	<0.0001	0.02
TIMP-2	24	24	24	33	0.003	n.s.
Endostatin	184	198	182	201	n.s.	n.s.
P-selectin	56	67	63	109	0.0003	0.04
ICAM-1	276	315	266	298	n.s.	n.s.
VCAM-1	135	136	141	144	n.s.	n.s.
Tie-2	116	162	129	375	0.005	n.s.
PAI-1	18	24	21	38	0.003	n.s.
MIF	53	55	51	78	n.s.	n.s.
uPAR	20	27	22	53	0.01	n.s.
Ang-2	978	1458	1447	2914	0.007	0.03
Follistatin	778	953	928	1704	0.006	n.s.
HGF	1933	2932	2803	5342	0.01	0.04
IL-8	38	56	53	108	0.002	0.02
G-CSF	595	1114	797	1132	n.s.	n.s.
PDGF-BB	442	822	802	1483	0.02	0.03
VEGF	129	209	192	375	0.006	0.03
Leptin	3236	2815	3235	3752	n.s.	n.s.
PECAM-1	5026	5914	5625	6356	n.s.	n.s.
IL-12	3	8	5	5	n.s.	n.s.
FGF-basic	19	32	16	28	n.s.	n.s.
TNF- α	4	6	3	2	n.s.	n.s.

Values indicate the average. Univariate: linear trend analysis, multivariate: proportional odds model. The concentrations are ng/mL for interleukin 6 receptor (IL-6R), matrix metalloproteinase 9 (MMP-9), TIMP metalloproteinase inhibitor 1 (TIMP-1), TIMP-2, endostatin, P-selectin, intercellular adhesion molecule 1 (ICAM-1), vascular cell adhesion molecule 1 (VCAM-1), Tie-2, plasminogen activator inhibitor-1 (PAI-1), macrophage migration inhibitory factor (MIF), and plasminogen activator urokinase receptor (uPAR), and the others are pg/mL. FGF-basic, fibroblast growth factor 2; G-CSF, colony stimulating factor 3; HGF, hepatocyte growth factor; n.s., not significant; PDGF-BB, platelet-derived growth factor beta polypeptide; PECAM-1, platelet/endothelial cell adhesion molecule; TNF- α , tumor necrosis factor; VEGF, vascular endothelial growth factor.

the recurrence (+) versus the recurrence (−) groups ($P = 0.335$) in our study.

Discussion

Vascular cell adhesion molecule 1 (VCAM-1)/CD106 is a member of the Ig superfamily and encodes a cell surface sialoglycoprotein expressed by cytokine-activated endothelium. This type I membrane protein mediates leukocyte-endothelial cell adhesion and signal transduction, and may play a role in the development of atherosclerosis⁽¹²⁾ and rheumatoid arthritis.⁽¹³⁾ In the field of oncology, accumulating evidence suggests that VCAM-1 is associated with a poor outcome.^(14–17) Recently, Shariat *et al.*⁽¹⁸⁾ reported that standard clinical variables alone exhibited an accuracy of 71.6% for predicting the risk of biochemical recurrence following a radical prostatectomy in patients with prostate cancer, whereas the addition of preoperative blood levels of TGF- β 1, sIL-6R, IL-6, VCAM-1, VEGF, endoglin, and uPA increased the predictive accuracy by 15–86.6%. Vascular endothelial growth factor (VEGF) and uPA were not significant predictors of post-operative recurrence in our data set for CRC, but VCAM-1 and sIL-6R were significant, consistent with Shariat’s study. The mechanism by which VCAM-1 mediates an unfavorable phenotype remains unclear, but the most probable explanation is that the tumor cells escape T-cell immunity by

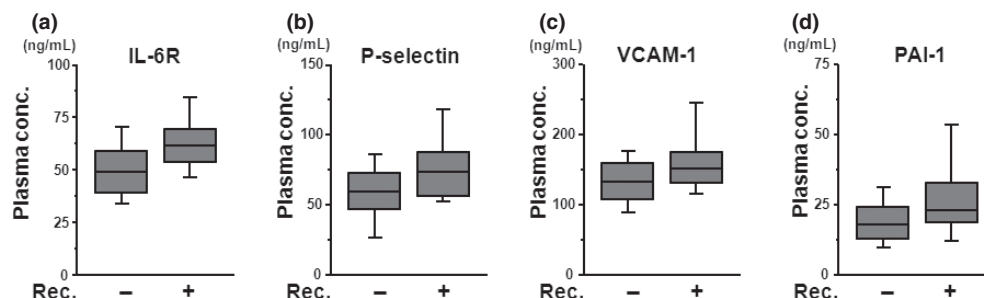


Fig. 1. The concentrations of (a) interleukin 6 receptor (IL-6R), (b) P-selectin, (c) vascular cell adhesion molecule 1 (VCAM-1), and (d) plasminogen activator inhibitor-1 (PAI-1) were significantly higher in the recurrence group in colorectal cancer. The upper bar, box, and lower bar represent the 90%, 75%, 50%, 25% and 10% percentiles. The plasma concentrations of each molecule were measured using a Bio-Plex suspension array system. Rec, recurrence.

Table 4. Results of multivariate analysis for recurrence after curative surgery in 96 colorectal cancers

Molecules	Recurrence*		Univariate		Multivariate
	+	-	t-test	Logistic	Stepwise
			P-value	P-value	P-value
IL-6R	63	51	0.03	n.s.	
MMP-9	37	35	n.s.	n.s.	
TIMP-1	119	113	n.s.	n.s.	
TIMP-2	25	24	n.s.	n.s.	
Endostatin	193	186	n.s.	n.s.	
P-selectin	76	60	0.04	n.s.	
ICAM-1	285	281	n.s.	n.s.	
VCAM-1	164	135	0.02	0.039	0.009
Tie-2	183	127	n.s.	n.s.	
PAI-1	28.3	19.8	0.02	0.005	0.005
MIF	64	51	n.s.	n.s.	
uPAR	30	21	n.s.	n.s.	
Ang-2	1514	1292	n.s.	n.s.	
Follistatin	962	883	n.s.	n.s.	
HGF	3155	2517	n.s.	n.s.	
IL-8	53	49	n.s.	n.s.	
G-CSF	892	810	n.s.	n.s.	
PDGF-BB	972	671	n.s.	n.s.	
VEGF	211	174	n.s.	n.s.	
Leptin	2623	3196	n.s.	n.s.	
PECAM-1	6159	5447	n.s.	n.s.	
IL-12	4	5	n.s.	n.s.	
FGF-basic	16	22	n.s.	n.s.	
TNF- α	4	4	n.s.	n.s.	

*Values indicate the average. Recurrence, post-operative recurrence; logistic, logistic regression model. The concentrations are ng/mL for interleukin 6 receptor (IL-6R), matrix metalloproteinase 9 (MMP-9), TIMP metalloproteinase inhibitor 1 (TIMP-1), TIMP-2, endostatin, P-selectin, intercellular adhesion molecule 1 (ICAM-1), vascular cell adhesion molecule 1 (VCAM-1), Tie-2, plasminogen activator inhibitor-1 (PAI-1), macrophage migration inhibitory factor (MIF), and plasminogen activator urokinase receptor (uPAR), and the others are pg/mL. FGF-basic, fibroblast growth factor 2; G-CSF, colony stimulating factor 3; HGF, hepatocyte growth factor; n.s., not significant; PDGF-BB, platelet-derived growth factor beta polypeptide; PECAM-1, platelet/endothelial cell adhesion molecule; TNF- α , tumor necrosis factor; VEGF, vascular endothelial growth factor.

overexpressing the endothelial cell adhesion molecule VCAM-1, which normally mediates leukocyte extravasation to sites of tissue inflammation.⁽¹⁹⁾

Plasminogen activator inhibitor-1 (PAI-1)/SERPINE1 belongs to the plasmin/plasminogen system and is secreted into the blood, where it prevents the generation of plasmin,

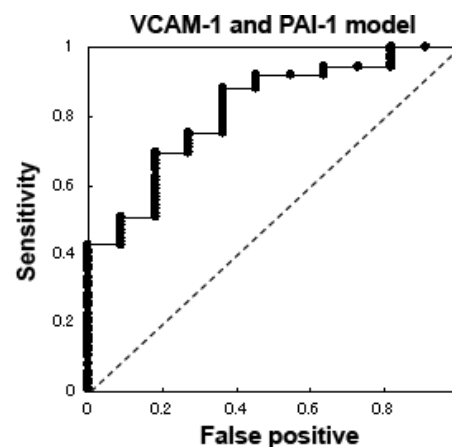


Fig. 2. The receiver-operator curve (ROC) for post-operative recurrence. A stepwise method selected vascular cell adhesion molecule 1 (VCAM-1) and plasminogen activator inhibitor-1 (PAI-1) as the most useful explanatory parameters; these two molecules were subsequently used to construct a prediction model. The ROC indicates the results of this model, which was capable of discriminating post-operative recurrence with an area under the curve (AUC) of 0.82, a sensitivity of 0.75, and a specificity of 0.73.

promoting the persistence and expansion of thrombi.⁽²⁰⁾ Plasminogen activator inhibitor-1 (PAI-1) is known as tumor biological prognostic factor and has been thoroughly validated with regard to its clinical utility in breast cancer.^(21,22) The 2007 Breast Tumor Markers Guidelines recommend that uPA/PAI-1 be measured using an ELISA with a minimum of 300 mg of fresh or frozen breast cancer tissue for determining the prognosis of patients with newly diagnosed, node-negative breast cancer. Furthermore, CMF-based adjuvant chemotherapy provides a substantial benefit, compared with observation alone, in patients with a high risk of recurrence as determined by the presence of high levels of uPA and PAI-1.⁽²³⁾ Previous reports have demonstrated that higher levels of PAI-1, but not PAI-2, are associated with large tumors, metastatic stage, and a worse prognosis in patients with CRC.⁽²⁴⁻²⁶⁾ Our study differed in that it evaluated the clinical parameter of post-operative recurrence in a prospective study. The biological mechanism by which PAI-1 promotes tumor progression is thought to involve a reduction in cell adhesion to the extracellular matrix as a consequence of excess PAI-1 interfering with uPAR binding to vitronectin, thereby facilitating cell invasion and migration.⁽²⁷⁾ Interestingly, accumulating data indicate that both VCAM-1 and PAI-1 promote tumor metastasis and cellular adhesion. These activities are likely involved in post-operative recurrence. We plan to perform

a validation study to predict post-operative recurrence using the plasma concentrations of VCAM-1 and PAI-1 in the near future.

In conclusion, we have demonstrated that a combination prediction model based on the plasma concentrations of VCAM-1 and PAI-1 was a useful biomarker for predicting post-operative recurrence in patients with colorectal cancer. Our strategy, which utilizes a multiplex immunoassay system, may be a powerful tool for identifying biomarkers in clinical settings.

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Disclosure Statement

The authors have no conflict of interest.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Fig. S1. Metastasis (+) vs (–) in independent samples of colorectal cancer (CRC) ($n = 28$).

Fig. S2. Post-operative recurrence (+) vs (–) in stage III colorectal cancer (CRC) ($n = 45$).

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